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INFORMATIONSKIOSK

Large scale RAG system in action

From RAG Prototype to Multi-Tenant Chatbot

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What was the need?

For Units



- **Fewer repetitive questions.**
- Clearer visibility into content gaps in their sources of information

For Users



- Faster access to relevant information
- Personalised queries
- **24 x 7** accessible

For University

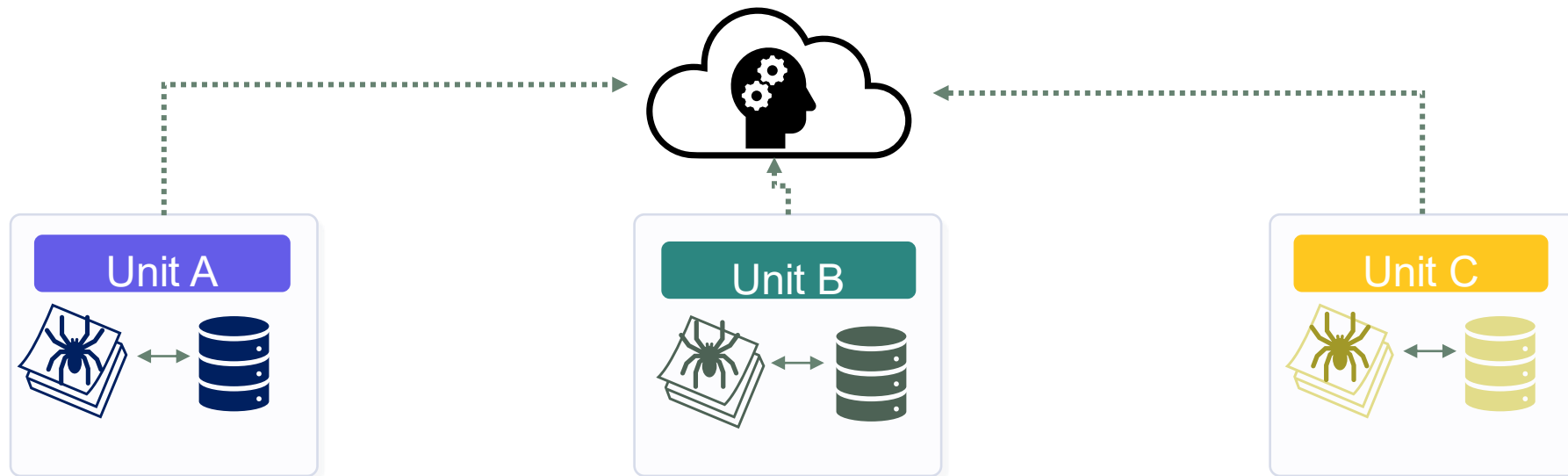


- **A reusable AI infrastructure instead of fragmented isolated chatbot pilots**
- Control over data management.
- **In House Solution and Innovation**

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What were we tasked to build?

- Multi-Tenant Chatbot Infrastructure
 - Serving multiple units simultaneously without impacting performance.
 - Supporting knowledge base and prompt isolation per unit.
 - With fully automated and scheduled pipeline for scraping and index formation for each unit.

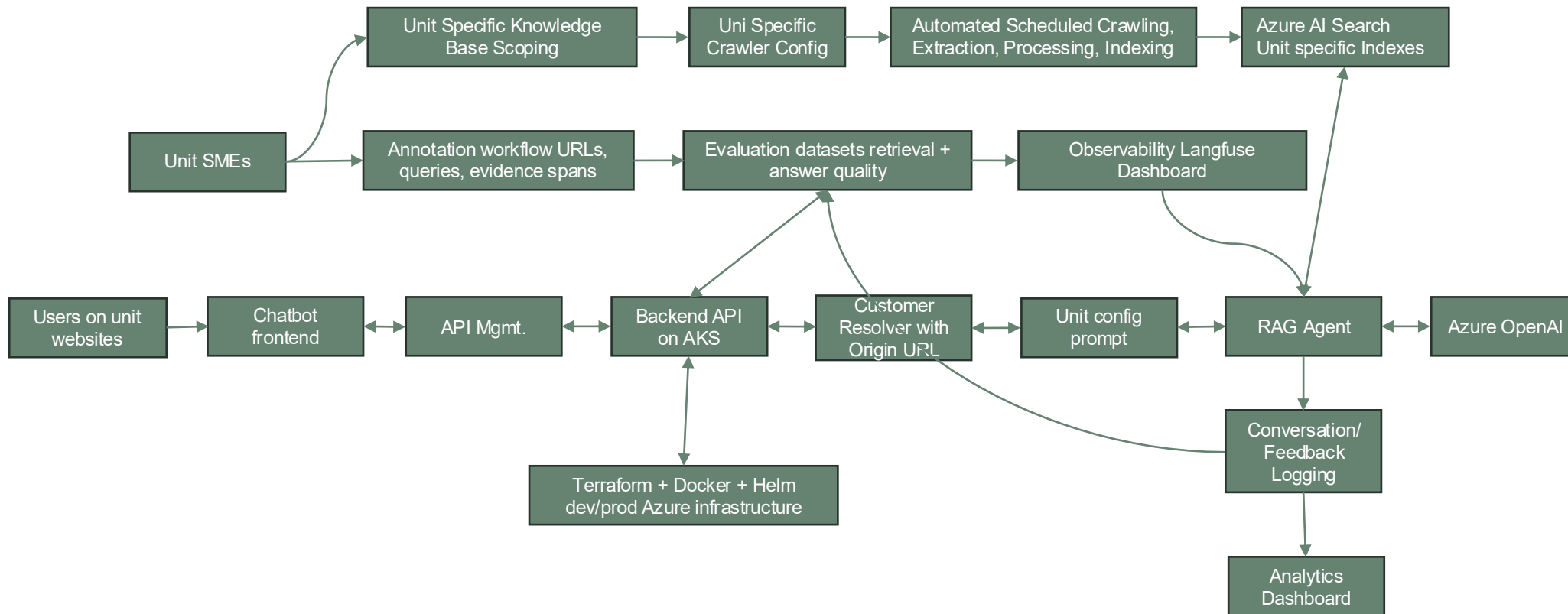


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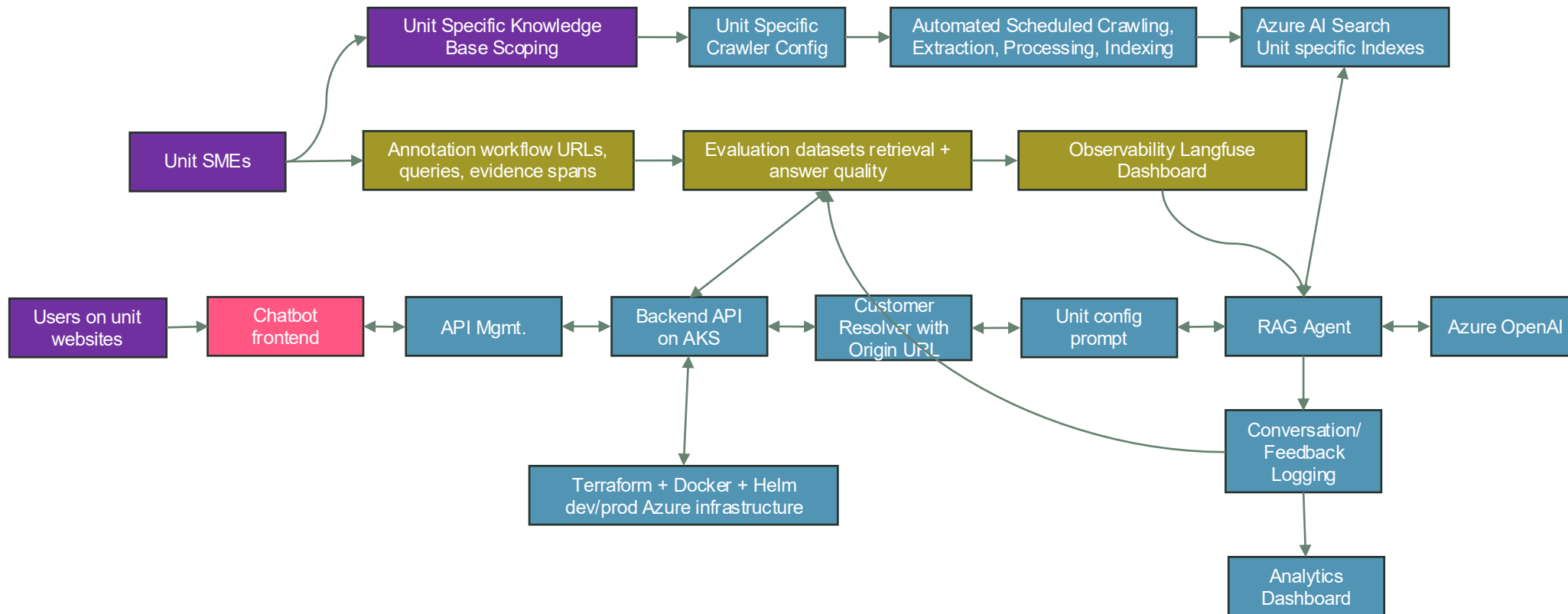
Project in Numbers

Metric	Value
Time period	July 2025 – June 2026
Capacity	~1 FTE
Repositories / major modules	Backend, ETL, Annotation, Dashboard, Evaluation, Frontend
Major workstreams	UI, RAG, ETL, Azure infra, deployment, testing, logging, annotation, release process

Current Workflow Overview

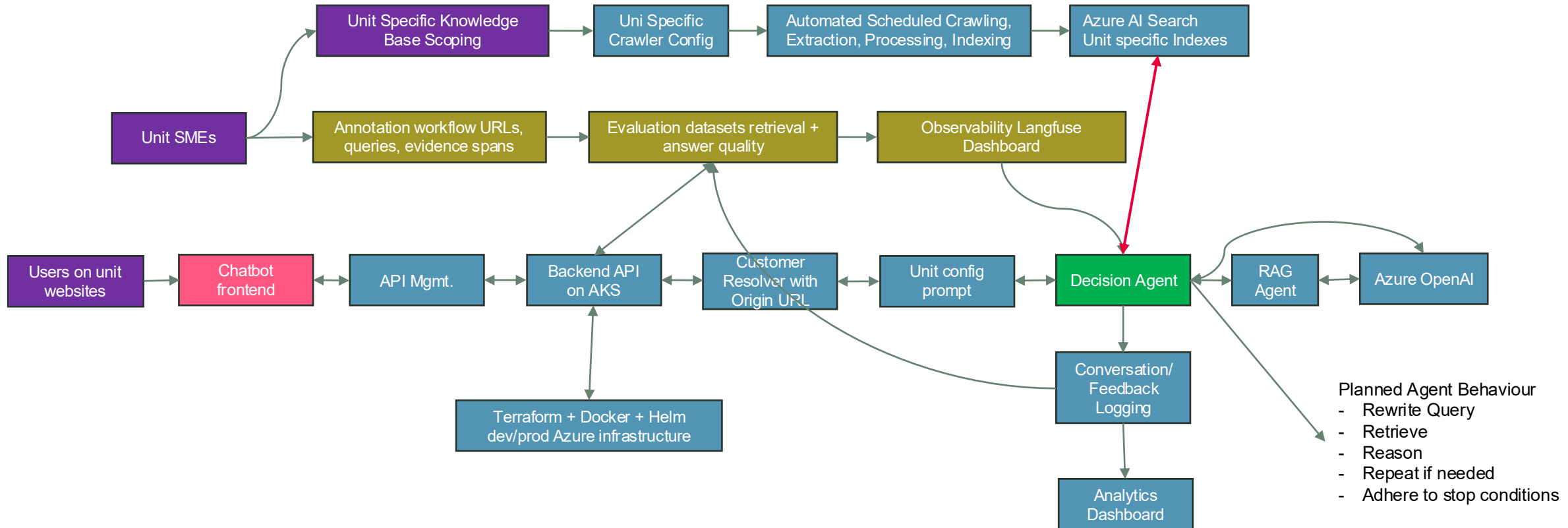


Current Workflow Overview



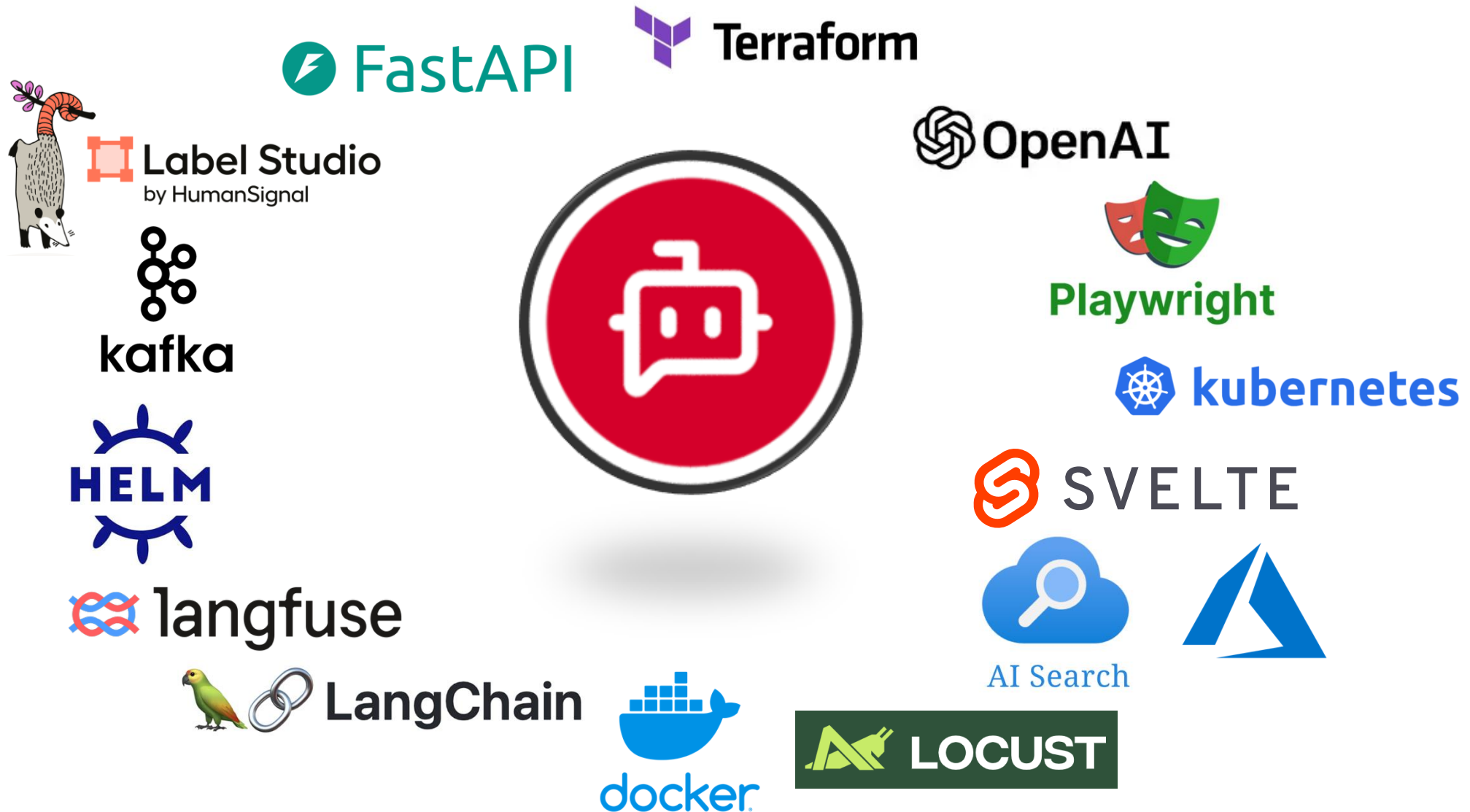
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Introducing Decision Agent (Ongoing)



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Technologies



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Application Code



SVELTE

Frontend Components

- Based on Svelte (Compiler based UI framework)
- Hosted on Github Pages



FastAPI



LangChain



OpenAI



AI Search



kafka

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Application Code



AI Search



Backend Components

- Web framework for building Python APIs
- Used in this project to serve the following endpoints
 - [/initialize-agent](#)
 - Initializes the agent with a specific [session_id](#).
 - Resolve unit specific knowledge base with [origin_url](#).
 - [/invoke-agent](#)
 - Retrieves from AI search given a user query.
 - Generates the answer.
 - Logs timestamped conversations with [session_id](#).
 - [/send-feedback](#)
 - Logs timestamped feedback for the user [session_id](#).

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Application Code



AI Search



Backend Components

- Python framework for building LLM powered applications
- Used in this project for
 - Formatting documents for LLM ingestion.
 - Prompt templates and chaining of prompts.
 - Formatting message objects and preserving conversation history.
 - Powering the agent and tools.

u^b Application Code



AI Search



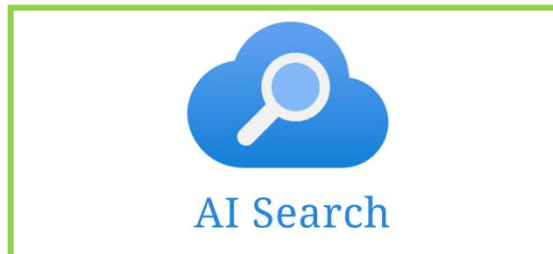
Backend Components

- Azure OpenAI LLM Provider
- Used in this project for
 - Powering the Agent (currently GPT-5.2).
 - Embedding the documents and query.
 - Translation.



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Application Code



Backend Components

- Azure AI search platform
- Used in this project to
 - Power the RAG.
 - Create unit specific indexes.
 - Hybrid (keyword + vector) queries.
 - Semantic Chunking.
 - Add AI generated Metadata to Chunks (Titles, Keywords + Questions)



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Application Code



AI Search



Backend Components

- Apache Kafka is a distributed event streaming platform for data processing.
- Used in this project
 - For logging chat interactions and feedback via event streaming.
 - As a broker & dedicated consumer service that persists logs into Azure Blob Storage for later analysis, auditing, or dataset generation.

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Deployment



Terraform



docker



kubernetes



Deployment Components



- Infrastructure as a code tool to provision, change and version infra safely and efficiently
- Used in this project for
 - Creating modules for management of all resources including
 - AKS cluster
 - Azure OpenAI
 - Azure AI Search
 - Storage Account
 - Container Registry
 - Key Vault
 - Networking
 - API Management
 - Managing separate deployment environments (dev, prod).
 - Bridging infrastructure and application configuration

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Deployment



Deployment Components

- Docker is used in this project for containerizing, standardizing and shipping the backend images for dev/prod, ETL pipeline, dashboard, annotation tool .

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Deployment



Deployment Components



- AKS simplifies the deployment, management, and operations of Kubernetes clusters
- Used in this project for
 - Hosting the backend API and the Kafka consumer, using the Kubernetes manifests under helm/templates.
- It sits in the Terraform-managed Azure infrastructure so the team can provision the cluster, connect it to the container registry, and (re)deploy the services

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Testing and Release Discipline

 langfuse



AI Search



Label Studio
by HumanSignal



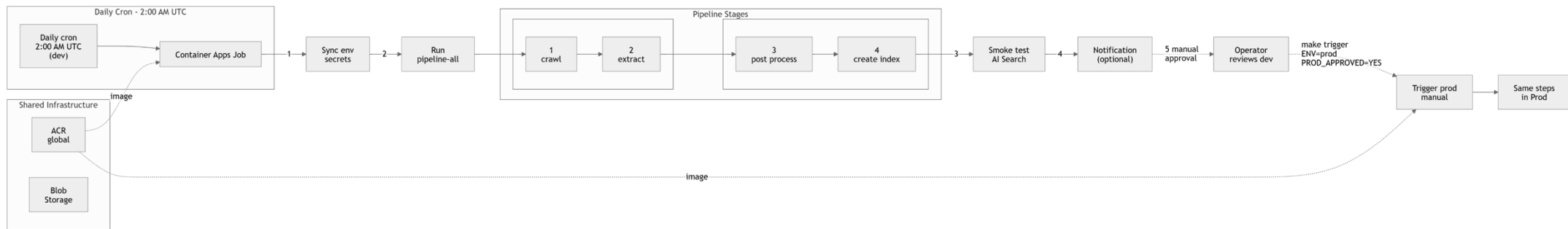
Playwright



Test type	Purpose
Unit tests	Agent components and expected behavior
E2E tests	Frontend/backend contract tests
Customer-specific tests	Ensure multi-unit routing works
Load tests	Rate limits and scaling behavior
Release process	Freeze, test, deploy, verify, release notes

ETL Pipeline

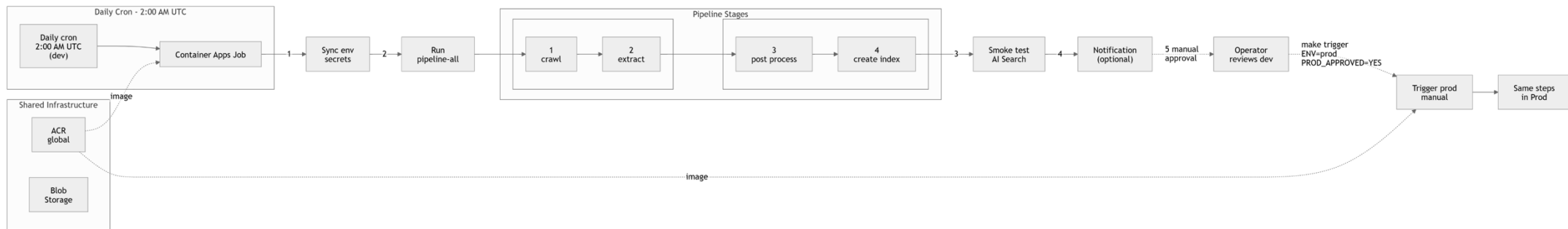
Making data refresh operational



- ETL Pipeline is applied separately for each unit.
 - Crawl step discovers the pages, pdfs based on unit configuration.
 - Extract step extracts the text from html/ pdfs.
 - Post process step normalizes and creates document level meta data.
 - Index step creates semantic chunks, generates chunk titles, archives the current index and replaces it
- In the future
 - Quality gates to based on evaluations will be implemented.
 - More meta data can be collected, and the cost of indexing can be lowered if direct access to search system of the CMS is allowed.

ETL Pipeline

Making data refresh operational



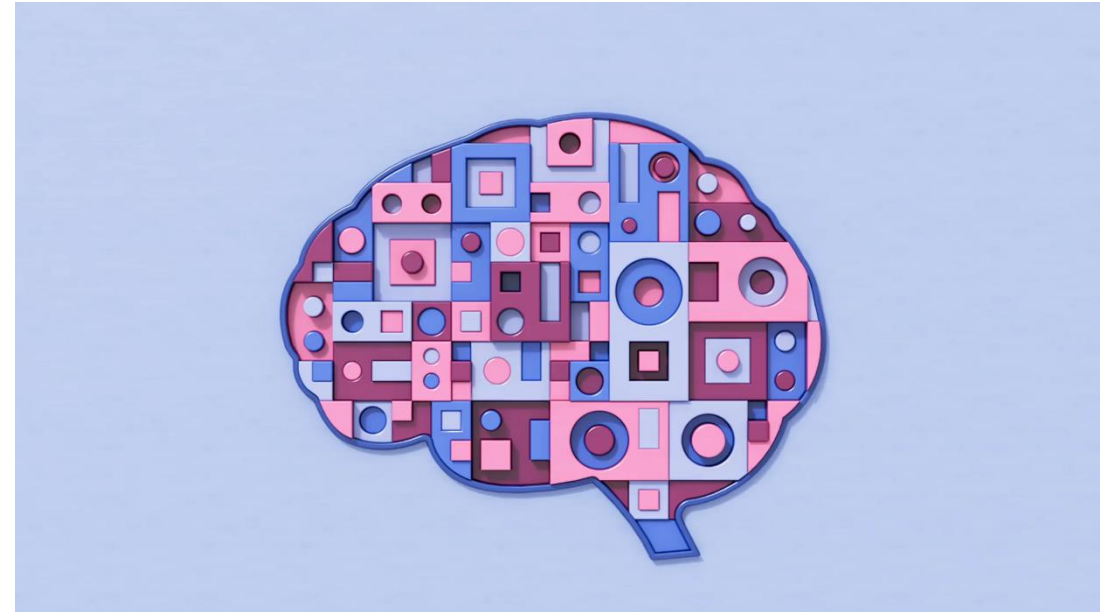
The catch:

*~ Freshness without validation can break trust.
Validation without automation does not scale.*

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What the chatbot **actually** has to decide?

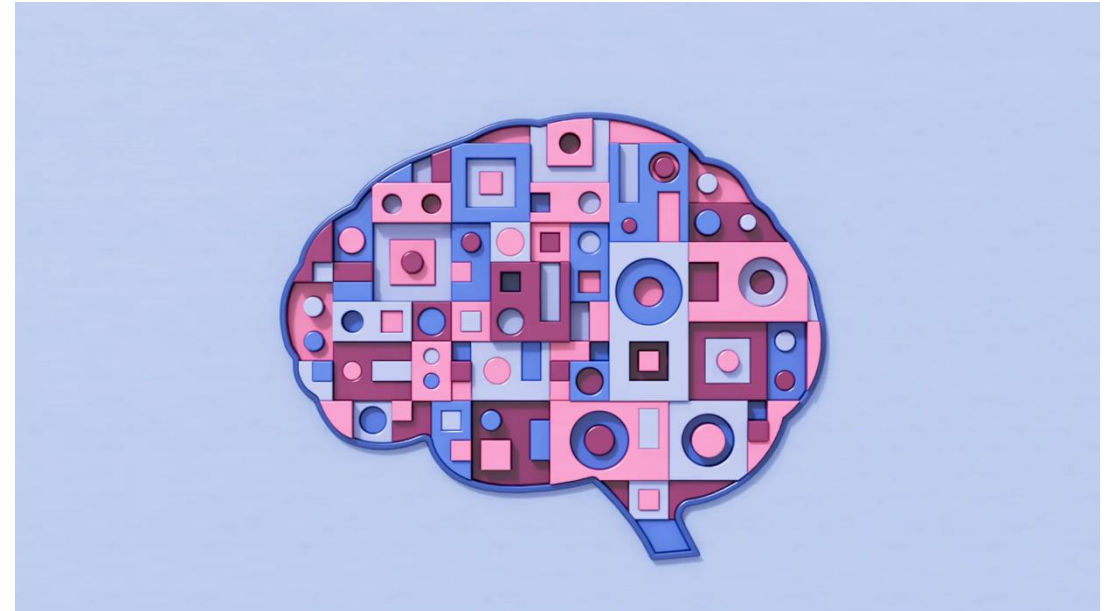
1. Is the question in scope?
2. Which unit does it belong to?
3. Should retrieval be used?
4. Which documents are relevant?
5. Is the evidence sufficient for answering?
6. Should it answer, clarify, refuse, or route to a contact?
7. Can it cite sources?
8. Should it preserve the user's language?
9. Should it modify query and re-retrieve?



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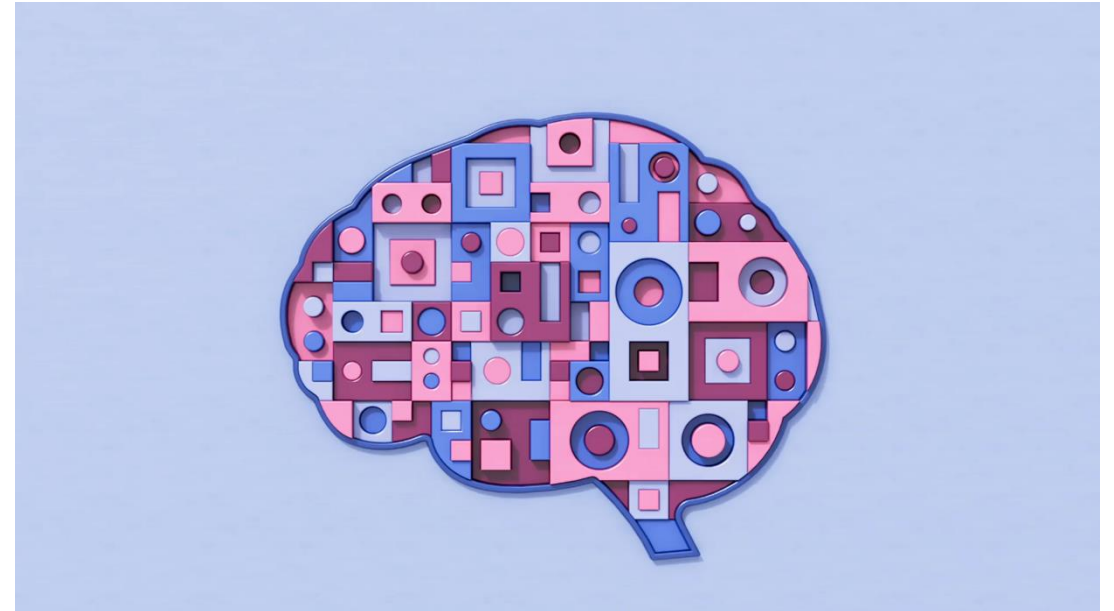
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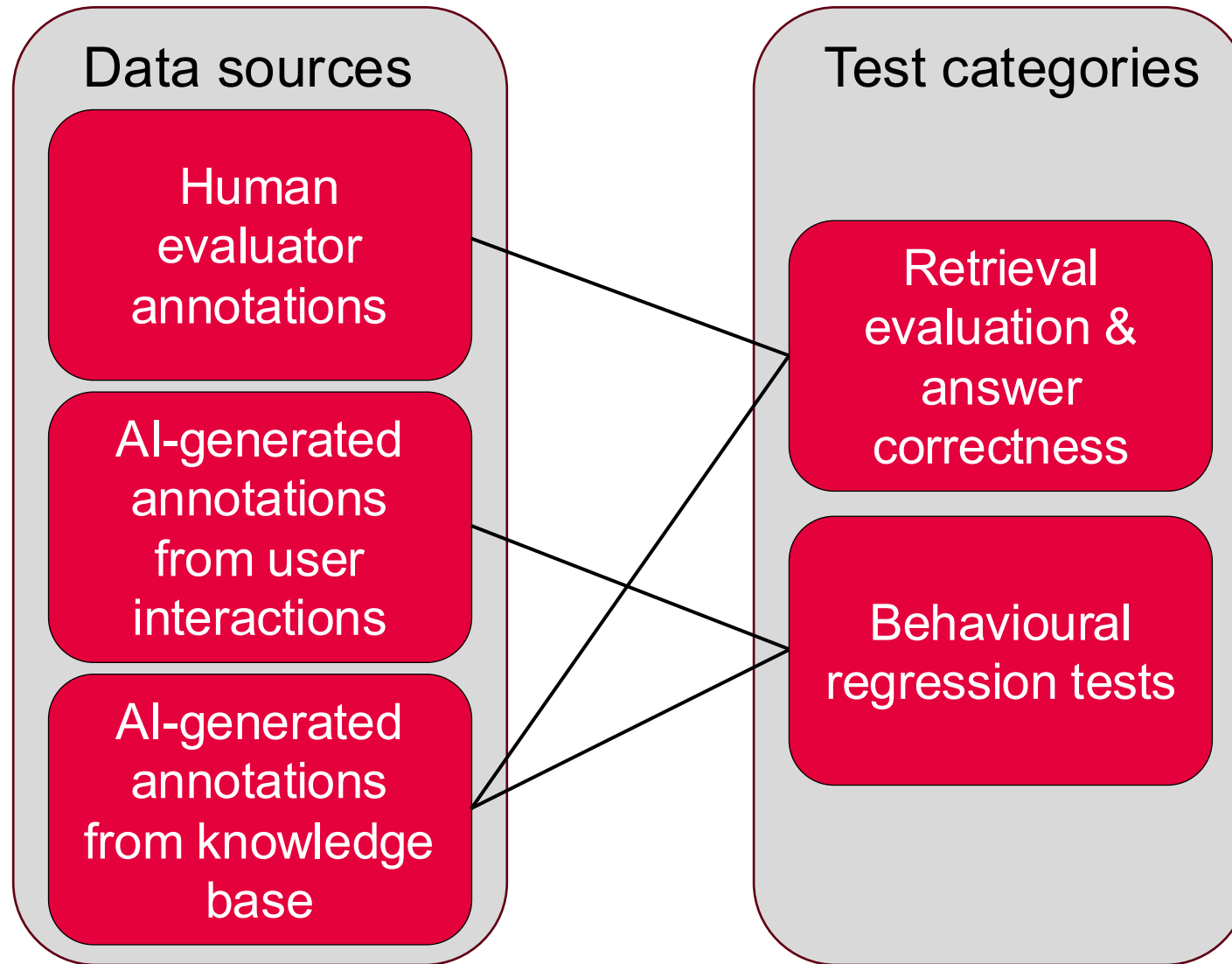
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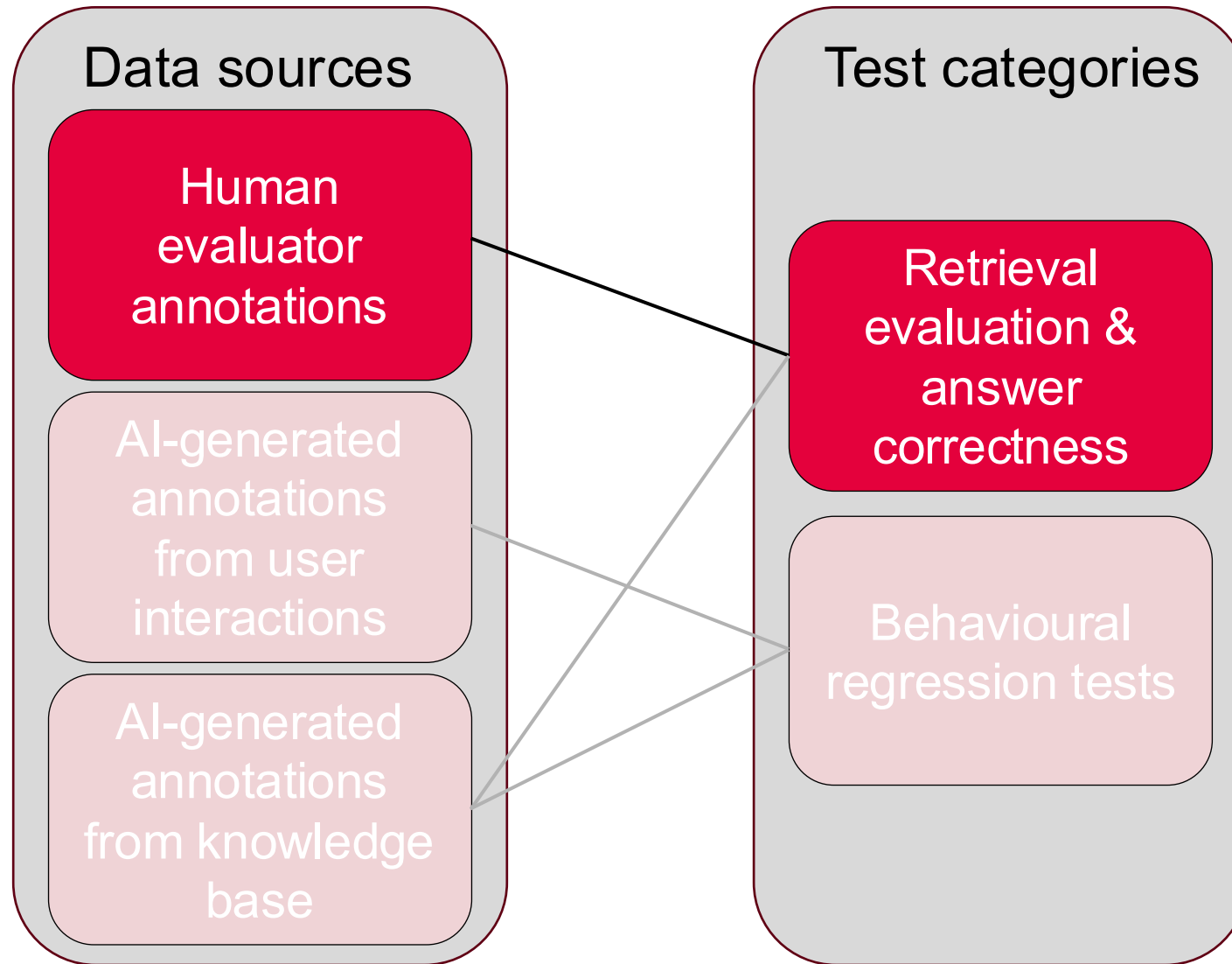
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Overview of Evaluation Strategy



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Overview of Evaluation Strategy



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Retrieval evaluation

Human annotations via Label Studio

Self-hosted Label Studio on Azure VM,
Docker deployment



Clients



Label Studio
by HumanSignal

Human
annotations

- Query
- Reference answer
- Expected documents
- Expected text span

Label Studio Projects / sorted_url_package / Labeling

	query	expected_answer	
☐	0 Frage Wer verantwortet die Lehrveranstaltungsevaluati	Fakultäten verantworten die Evaluationspläne und	{{'v', e.ch
☐	0 Frage: Legen Sie die Lehrveranstaltungen im	Evaluationen werden mit dem neuen	{{'v', e.ch
☒	0 Was ist Evasys?	System über das die Evaluationen durchgeführt	{{'v', e.ch
☐	0 Wer leitet die Fachstelle Lehrevaluationen?	Mina Dimitrova, eventuell zusätzliche Infos zum	{{'v', e.ch
☐	0 Wie viele Mitarbeiter hat die Fachstelle	Mina, Magdalena, Nik und My-Yen, eventuell	{{'v', e.ch
☐	0 Was ist der Unterschied zwischen Lehrevaluation	Lehrevaluation ist breiter Begriff für Evaluation der	{{'v', e.ch
☐	0 Und was ist der Unterschied zwischen	Meines Erachtens ist Lehrevaluation der	{{'v', e.ch
☐	0 Was ist eine breite Beurteilung der Qualität in	Der Begriff war mir so nicht geläufig.	{{'v', e.ch
☐	0 Wie sieht eine Studienprogrammevaluatio	Etwas belanglose aber korrekte Antwort, die leider	{{'v', e.ch
☐	0 Welche Evaluationsinstrumente	Evaluation von Lehre, Forschung und	{{'v', e.ch
☐	0 Wie oft muss ich meine Lehrveranstaltung	Jede Veranstaltung muss einmal alle 3 Jahre evaluiert	{{'v', e.ch

Query

Was ist Evasys?

Expected answer (edit if needed)

System über das die Evaluationen durchgeführt werden. Bietet unter anderem automatische E-Mails und Auswertungen der Ergebnisse

127 characters Press Shift + Enter to Add + Add

Select relevant URLs (click a link to open in new tab)

- https://intern.unibe.ch/dienstleistungen/lehre/lehrevaluation/digitalisierung_evaluation/index_ger.html^[1]
- https://intern.unibe.ch/dienstleistungen/lehre/lehrevaluation/digitalisierung_evaluation/projektstatus/index_ger.html^[2]
- https://intern.unibe.ch/dienstleistungen/lehre/lehrevaluation/digitalisierung_evaluation/schulungen_und_unterlagen/index_ger.html^[3]
- <https://intern.unibe.ch/unibe/uniintern/content/e1883/e1945/e282399/e1493122/e1622946/e1644407/pane1644408/e1660366>
- <https://intern.unibe.ch/unibe/uniintern/content/e1883/e1945/e282399/e1493122/e1622946/e1644407/pane1644420/e1644600>

#692 Submit

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Retrieval evaluation

Human annotations: from Label Studio to Langfuse



Human
annotations

- Query
- Reference answer
- Expected documents
- Expected text span

export &
upload

Reference
dataset

- Set of queries
- Expected output

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Retrieval evaluation

Human annotations: from Label Studio to Langfuse



Human
annotations

export &
upload

Reference
dataset

API call

KioskBot

- Query
- Reference answer
- Expected documents
- Expected text span

- **Set of queries**
- Expected output

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Retrieval evaluation

Human annotations: from Label Studio to Langfuse



Clients



Human
annotations

export &
upload

Reference
dataset

Evaluation

KioskBot

- Query
- Reference answer
- Expected documents
- Expected text span

- Set of queries
- **Expected answer**
- **Expected documents**

- **Generated answer**
- **Retrieved documents**

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How can we evaluate retrieval?

Retrieval quality

- Did the KioskBot RAG retrieve the expected documents?
- Did the expected document appear in the top-k results?
- Did the retrieved chunks contain the expected evidence span?

Deterministic retrieval metrics (e.g. Recall@k, Precision@k)

- Used for exact matching against annotated references.

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Retrieval evaluation approach



RAGAS integration

RAGAS is an open-source LLM-based evaluation framework for RAG systems. Used to evaluate:

- semantic relevance of retrieved chunks
- usefulness of retrieval context for answering a query

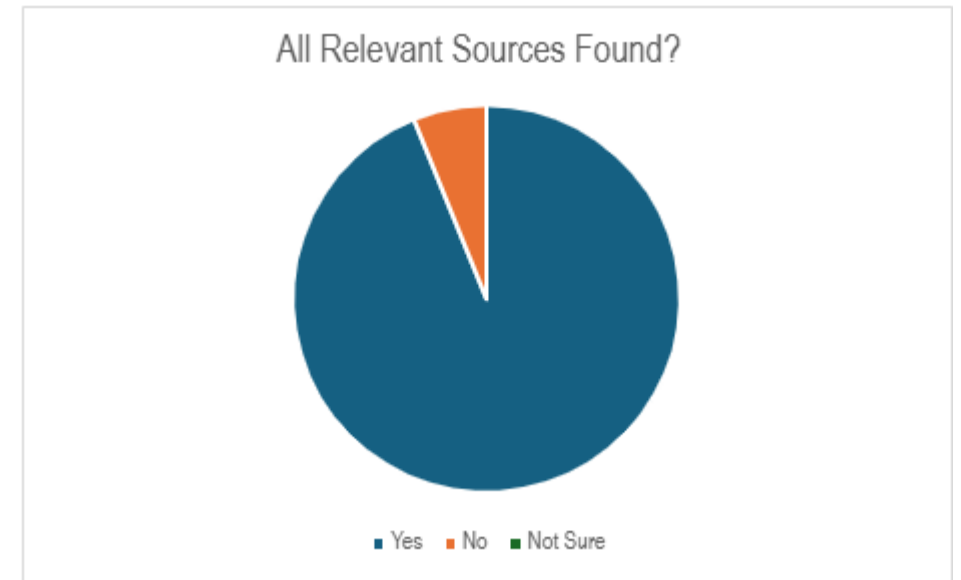
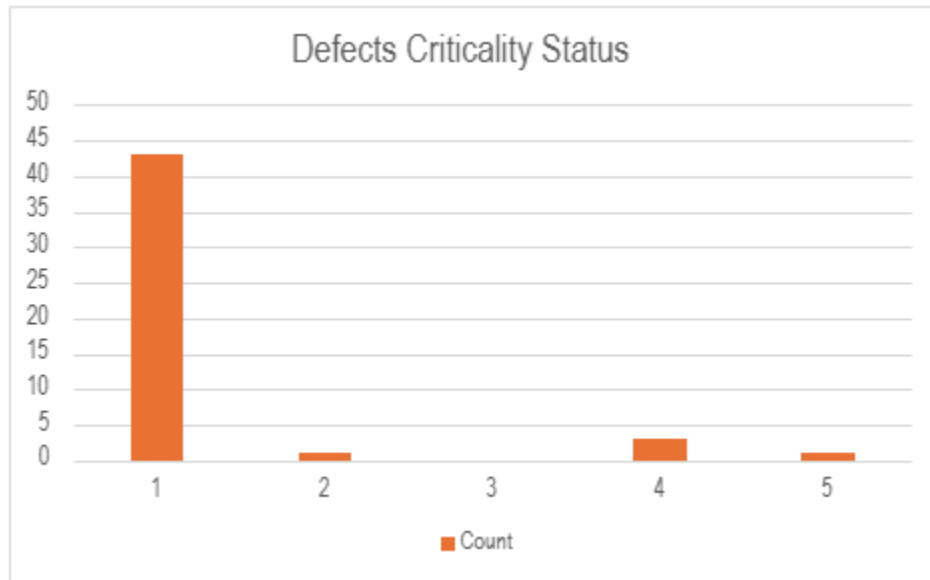
Example metrics:

- Context Precision@k
- Faithfulness
- Answer Relevancy
- Context Recall

Azure OpenAI API for LLM-as-a-judge

u^b Test Results (Innovation)

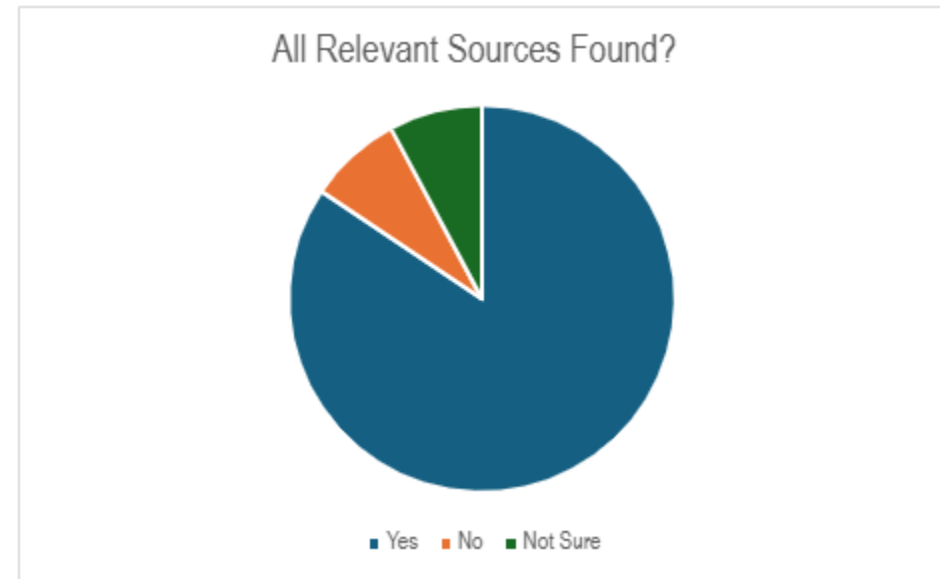
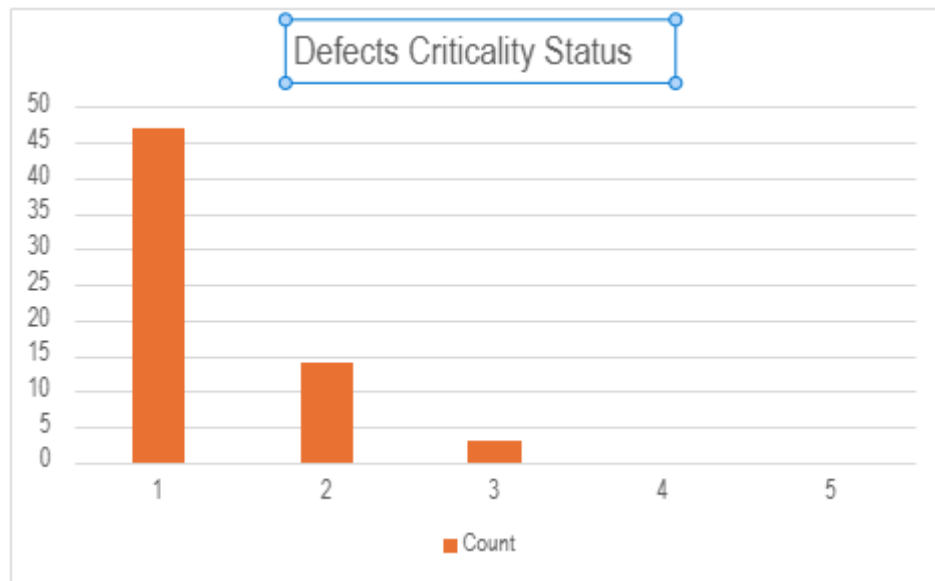
Correctness Accuracy ~ 0.89 (fully correct & human annotated)



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Test Results (QAD)

Correctness Accuracy ~ 0.73 (fully correct & human annotated)



u^b Informationskiosk Demo

Innovation Office



QAD Office (Quality Assurance and Development)



u^b Expectation vs Reality

Saving Manhours per unit

~ Effort vs Cost Tradeoff

- Estimate on a unit level how many hours spent in answering requests that could be automated as target cost. (X)
- Estimate unit specific effort needed for onboarding and maintenance as unit operational cost. (Y)
- Estimate what a solution costs per month per unit. (Z)
- **Goal** ~ $X > Z + Y$
 - Estimated avoided support effort > unit effort + solution cost
- Observe if goal is reached (moving average) over a given number of months.

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Key Learnings

Unfrozen requirements while flexible, delay development.

Assigning data ownership is important.

Website data needs to be crawler friendly.

There is a gap between Launch and Technology Adoption.

More evaluations and annotations lead to more product stability.

Multi-unit scaling requires process, not just code.

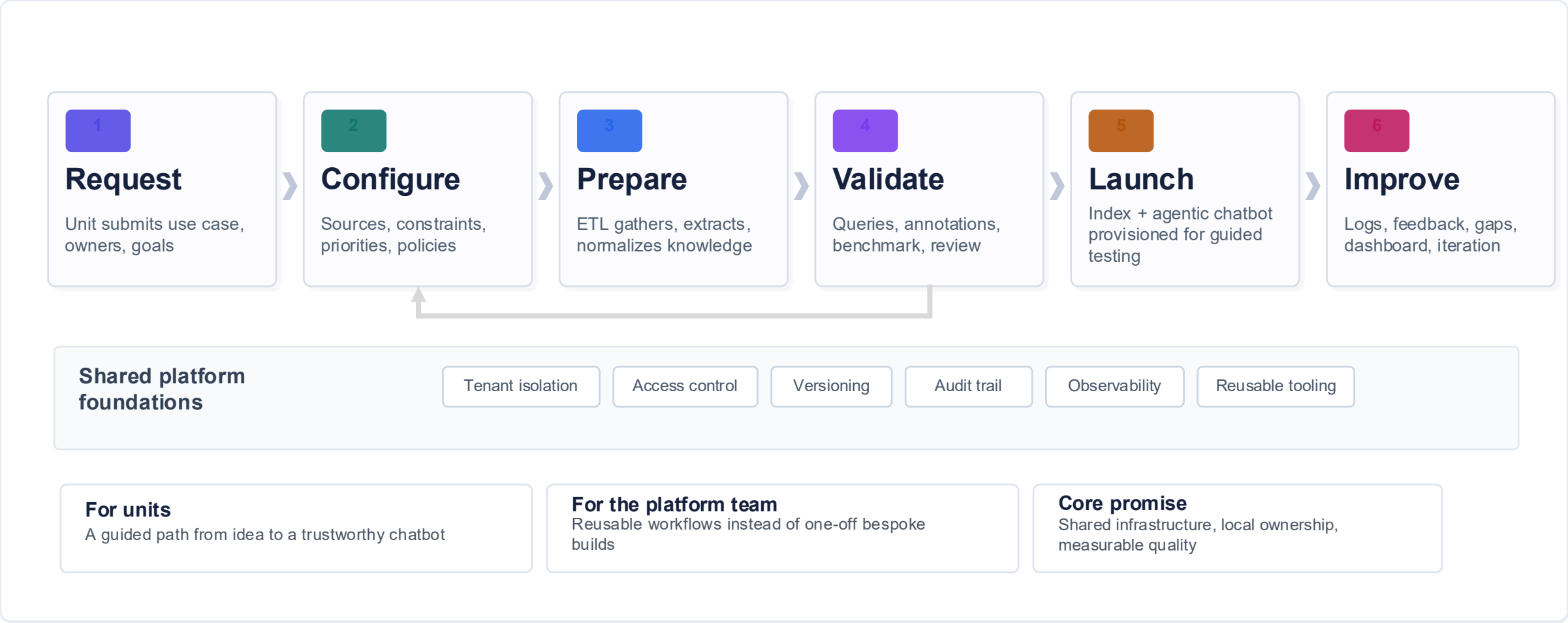
Bottleneck is not model capability but governance, labelling, and operational maintenance.

Estimating effort cost tradeoff to check expectations

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Guided Multi-Tenant Agentic Chatbot Platform (Future Vision)

A shared platform that helps each unit request, validate, launch, and continuously improve its own chatbot - with central guidance, shared infrastructure, and clear accountability.



u^b What support is needed?

Unit-side SMEs

For scoping, validation, representative questions, and launch approval.

PMO/process support

For requirements ownership, acceptance criteria, scheduling review cycles, communication.

Engineering capacity

For platformization, monitoring, security, evaluation, scaling.

Governance clarity

Who owns correctness? Who approves go-live? Who maintains content?

Sustainable funding model

Because onboarding and maintenance are recurring efforts in addition to new feature development.

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Dev Team Questions?

[Sukanya Nath](#) (Backend and Project Lead)

[Mariella Greutmann](#) (Evaluations)

[Sebastian Flick](#) (Frontend)

Paolo Rosso (Backend – Former)

